CapHive MCP Server — Team Testing Guide

A step-by-step guide for team members to test the CapHive MCP Server on staging with Claude.ai, ChatGPT, and Gemini.

What is This?

The CapHive MCP Server allows external AI clients (Claude.ai, ChatGPT, Gemini) to query fund management data directly from CapHive. Users authenticate via OAuth, and the AI can then call 23 read-only tools covering funds, capital activity, portfolios, cap tables, documents, and commitments.

Staging URL: `https://dev.altconnects.com/mcp`

Prerequisites

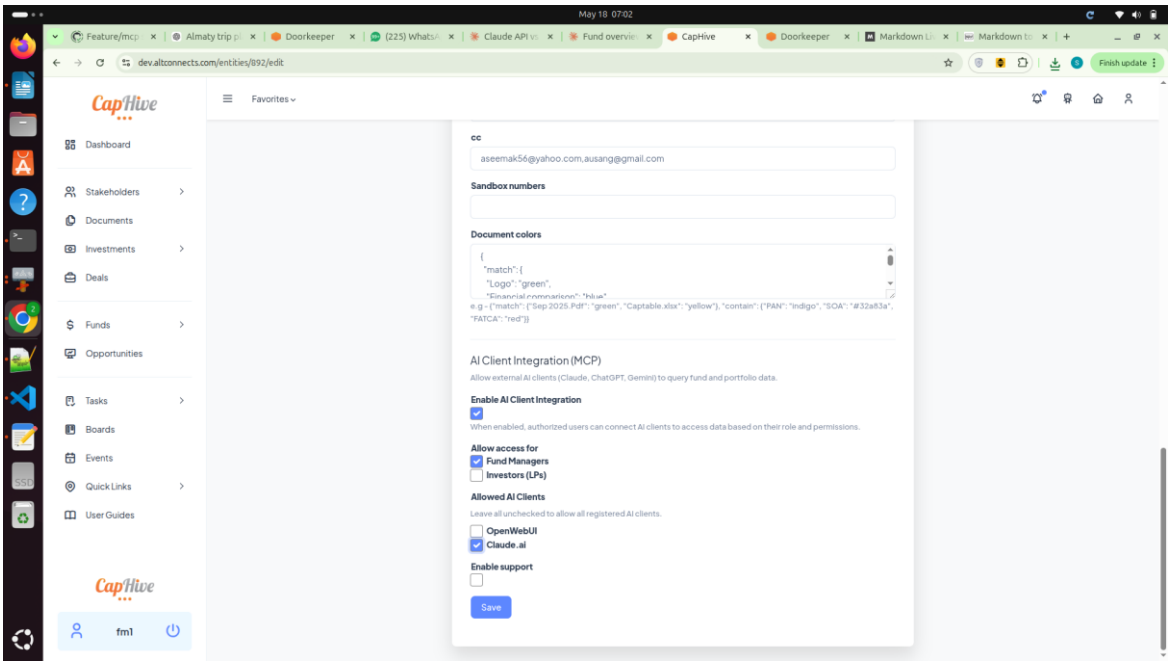
Before testing, an admin must complete the one-time server-side setup (Section A). Once that's done, each team member follows Section B to connect their AI client.

Section A: Admin Setup (One-Time)

A1. Enable MCP for Your Entity

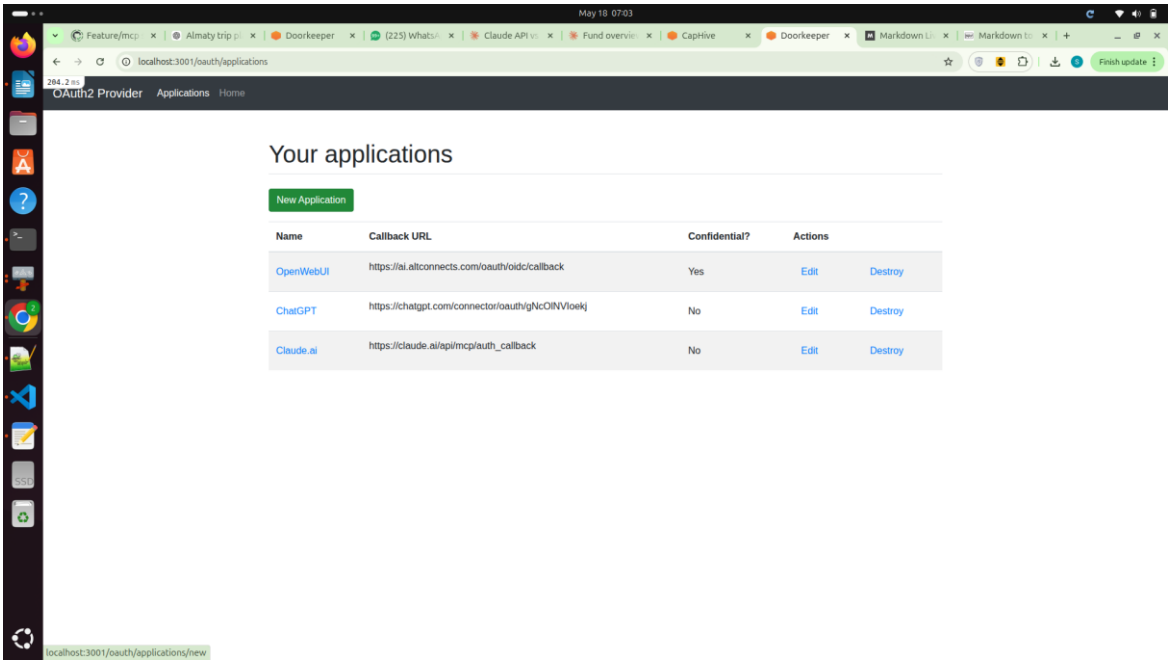
1. Log in to staging as an admin: `https://dev.altconnects.com`
2. Go to **Entity Settings** → Edit your entity
3. Scroll to the **"AI Client Integration (MCP)"** section
4. Check **"Enable AI Client Integration"**
5. Under **Allowed Roles**, check the roles that should have MCP access:
    - **Fund Managers** (employee role)
    - **Investors / LPs** (investor role)
6. Under **Allowed AI Clients**, leave all unchecked to allow all registered clients (or check specific ones to restrict)

7. Save



A2. Register OAuth Applications

Go to `**https://dev.altconnects.com/oauth/applications**` (you must be logged in).

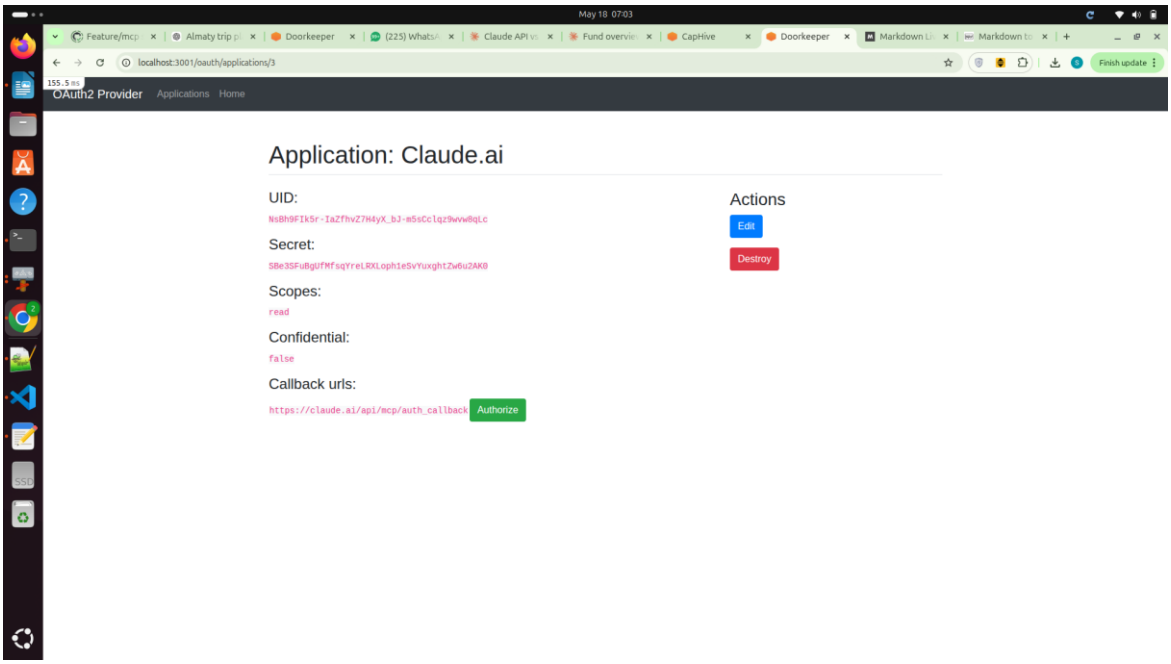


Click **"New Application"** and create an app for each AI client you want to test:

For Claude.ai

Field	Value
Name	`Claude.ai`
Redirect URI	`https://claude.ai/api/mcp/auth_callback`
Confidential	Unchecked (No)
Scopes	`read`

After saving, you'll see the **Client ID (UID)** and **Client Secret**. Note them down — though for Claude.ai the auto-discovery flow means you typically won't need to enter these manually.



For ChatGPT

Field	Value

Field	Value
Name	`ChatGPT`
Redirect URI	`https://chatgpt.com/connector/oauth/gNcOlNVloekj`
Confidential	Unchecked (No)
Scopes	`read`

> Note: The ChatGPT redirect URI contains a unique connector ID. You'll get this when setting up the connector in ChatGPT — see Section B2.

For Gemini

Field	Value
Name	`Gemini`
Redirect URI	*(TBD — depends on Gemini's MCP connector callback URL)*
Confidential	Unchecked (No)
Scopes	`read`

A3. Verify the Server is Accessible

Run this from your terminal to confirm the MCP endpoint is live:

```
curl -X POST https://dev.altconnects.com/mcp \
  -H "Content-Type: application/json" \
  -d '{"jsonrpc":"2.0","method":"initialize","id":1}'
```

Expected response:

```
{"error":"Missing token"}
```

This means the endpoint is live and auth is working (it correctly rejects unauthenticated requests).

To verify the OAuth discovery endpoints:

```
    # OAuth server metadata
    curl -s https://dev.altconnects.com/.well-known/oauth-
authorization-server | python3 -m json.tool

    # MCP resource metadata
    curl -s https://dev.altconnects.com/.well-known/oauth-protected-
resource | python3 -m json.tool
```

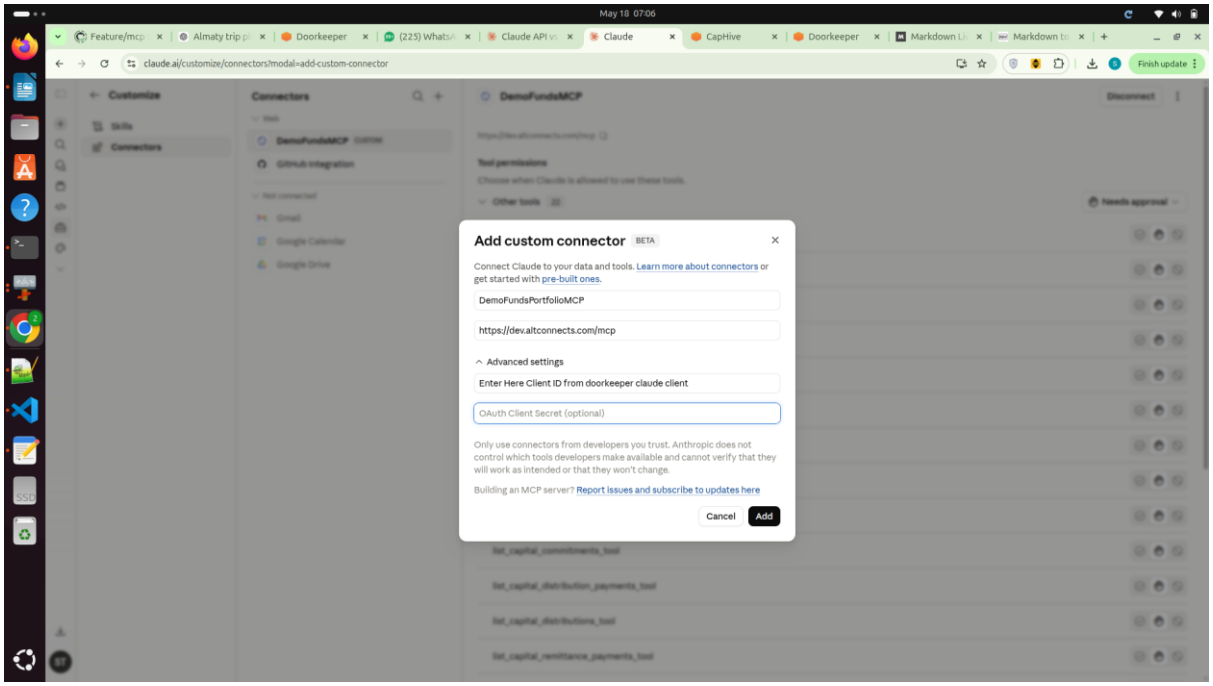
Both should return JSON with the OAuth endpoints and MCP resource info.

Section B: Connecting Your AI Client

B1. Claude.ai

Step 1: Add the CapHive Connector

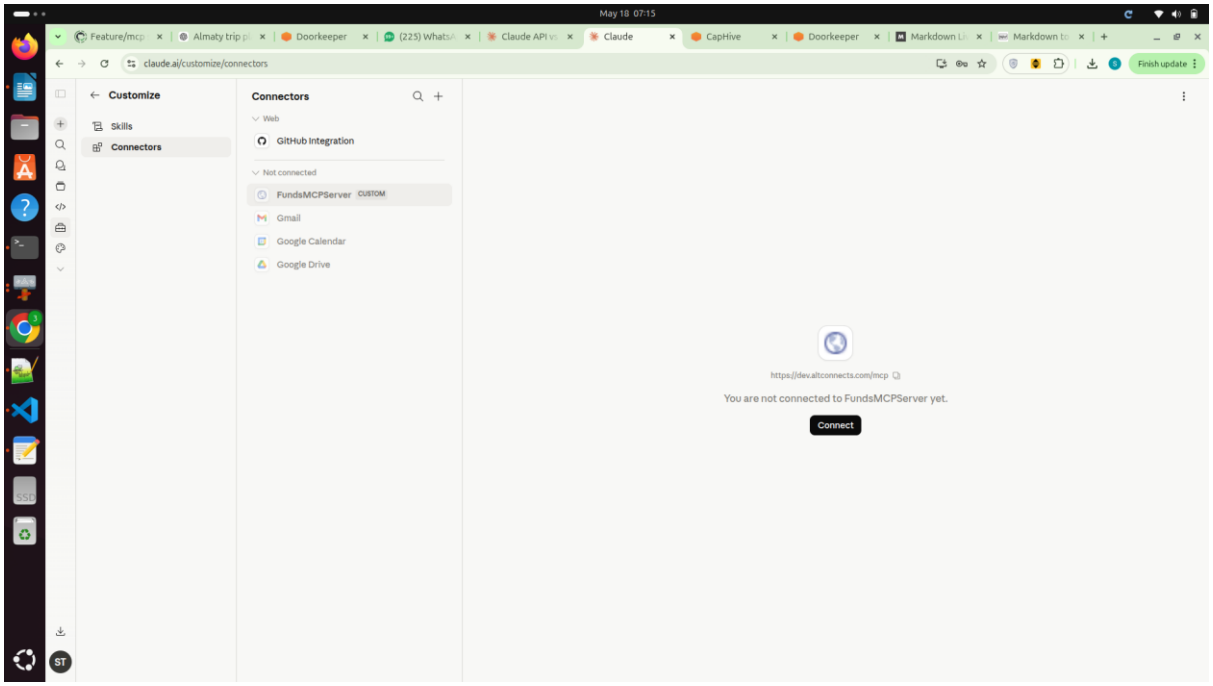
1. Go to claude.ai
2. Click **Settings** (gear icon) → **Customize** → **Connectors**
3. Click the "+" button to add a new connector
4. Enter the following:
    - **Name:** `CapHive` (or any name you prefer)
    - **URL:** `https://dev.altconnects.com/mcp`



5. Click **Save** / **Add**

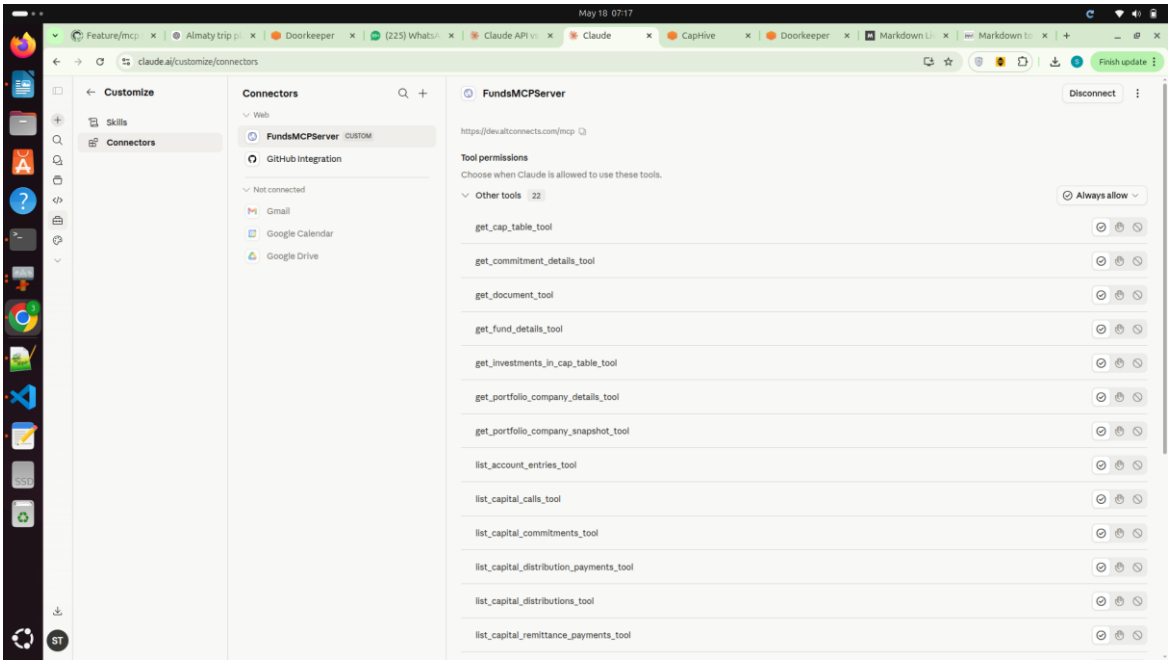
Step 2: Connect and Authorize

1. Click on the newly added **CapHive** connector
2. Click the **"Connect"** button



3. A browser popup opens with the CapHive login page
4. Log in with your staging credentials (e.g., `fm1@ian.com` / `password`)

5. You'll see an authorization prompt — click **"Authorize"**
6. The popup closes and you're connected. You should see a green "Connected" status.



Step 3: Test It

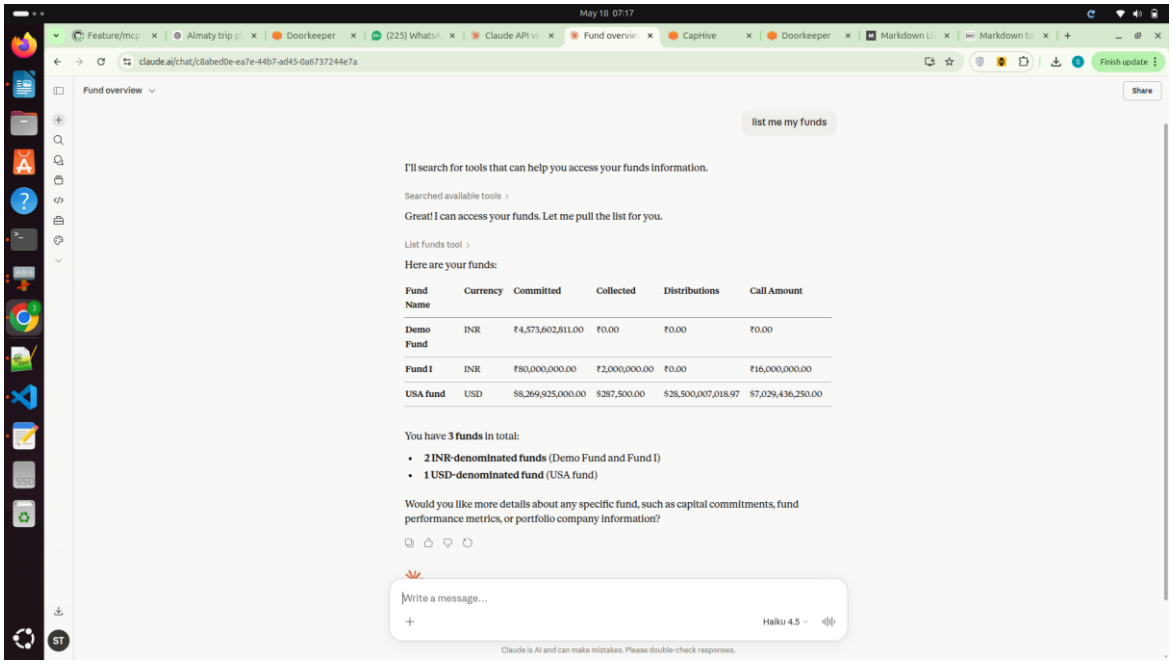
Start a new conversation on Claude.ai and try these queries:

```
List all my funds
```

```
Show me capital calls for Fund Alpha
```

```
What are the portfolio companies in my fund?
```

```
Show me the cap table for [company name]
```



B2. ChatGPT

Step 1: Add the CapHive Connector

1. Go to chatgpt.com
2. Click your profile → **Settings** → **Connected Apps** (or **Connectors**)
3. Click **"Add Connector"** → **"Custom"**
4. Enter the MCP server URL: `https://dev.altconnects.com/mcp`
5. ChatGPT will auto-discover the OAuth configuration from the `.well-known` endpoints

Step 2: Authorize

1. ChatGPT will redirect you to CapHive's login page
2. Log in with your staging credentials
3. Click **"Authorize"** on the consent screen
4. You'll be redirected back to ChatGPT

Step 3: Test It

Start a new conversation and try:

```
Using CapHive, list all my funds with their committed amounts
```

> Note: ChatGPT may require you to explicitly mention "CapHive" or select the connector when asking questions.

B3. Gemini

(Testing instructions will be added once Gemini MCP connector setup is confirmed.)

Available Tools (23 Total)

Once connected, the AI client can discover and call these tools:

Fund Tools

Tool	Description
`list_funds`	List all funds with committed, collected, distribution amounts
`get_fund_details`	Detailed info for a specific fund (terms, dates, currency)
`list_fund_ratios`	Fund performance ratios (TVPI, RVPI, DPI, IRR)

Capital Activity Tools

Tool	Description
`list_capital_calls`	Capital calls with amounts, due dates, status
`list_capital_commitments`	Investor commitments per fund

Tool	Description
`list_capital_distributions`	Distributions with gross amounts and dates
`list_capital_remittances`	Payments received and verification status
`list_capital_remittance_payments`	Individual remittance payment details
`list_capital_distribution_payments`	Individual distribution payment details

Portfolio Tools

Tool	Description
`list_portfolio_companies`	Portfolio company list with filtering
`get_portfolio_company_details`	Detailed company info
`get_portfolio_company_snapshot`	Company snapshot summary
`list_portfolio_investments`	Investments (buys, sells, FMV, gains)
`list_portfolio_kpis`	Company KPIs (actuals, budgets, periods)
`list_valuations`	Per-share valuations with dates

Cap Table Tools

Tool	Description
`get_cap_table`	Cap table for a portfolio company
`get_investments_in_cap_table`	Investments by investor, round, category

Document Tools

Tool	Description
`list_documents`	Document listing with filtering

Tool	Description
`get_document`	Document details and content

Commitment Tools

Tool	Description
`get_commitment_details`	Commitment details for a specific investor/fund
`list_fund_units`	Fund unit/class breakdown
`list_account_entries`	Account-level transaction entries

Sample Queries to Try

Here are some practical queries to test the full range of tools:

Fund Overview

- "List all my funds"
- "What is the total committed capital across all funds?"
- "Show me fund details for [fund name]"
- "What are the TVPI and IRR ratios for my funds?"

Capital Activity

- "Show me all capital calls due this month"
- "List the capital commitments for [fund name]"
- "What distributions have been made from [fund name]?"
- "Show remittance payments for the latest capital call"

Portfolio

- "List all portfolio companies"
- "Give me a snapshot of [company name]"
- "What investments have been made in [company name]?"

- "Show me the latest KPIs for [company name]"
- "What are the current valuations for [company name]?"

Documents & Cap Table

- "List recent documents"
- "Show me the cap table for [company name]"
- "Who are the investors in [company name]'s cap table?"

Cross-Domain Analysis

- "Compare the committed amounts across all my USD funds"
- "Which portfolio company has the highest FMV?"
- "Summarize the capital activity for [fund name] this quarter"

Troubleshooting

"Authorization with the MCP server failed"

Cause: MCP is not enabled for your entity, or your role is not in the allowed list.

Fix: Ask an admin to:

1. Go to Entity Settings → Edit
2. Enable "AI Client Integration"
3. Check your role under "Allowed Roles"

"Missing token" error

Cause: The AI client hasn't completed the OAuth flow yet.

Fix: Click "Connect" on the connector and complete the login/authorization flow.

"Invalid or expired token"

Cause: Your OAuth token has expired (tokens last 30 days).

Fix: Disconnect and reconnect the connector to get a fresh token.

"This AI client is not authorized for your organization"

Cause: The entity has restricted which AI clients can connect, and yours isn't in the list.

Fix: Ask an admin to add your AI client to the "Allowed AI Clients" list in Entity Settings, or uncheck all clients to allow all.

Connection works but no data returned

Cause: The user account may not have data in the staging environment, or the query doesn't match any records.

Fix: Try logging in with a user that has fund data (e.g., `fm1@ian.com` on staging).

OAuth login page doesn't appear

Cause: The `.well-known` metadata endpoints may not be configured correctly.

Fix: Verify by running:

```
curl -s https://dev.altconnects.com/.well-known/oauth-authorization-server | python3 -m json.tool
```

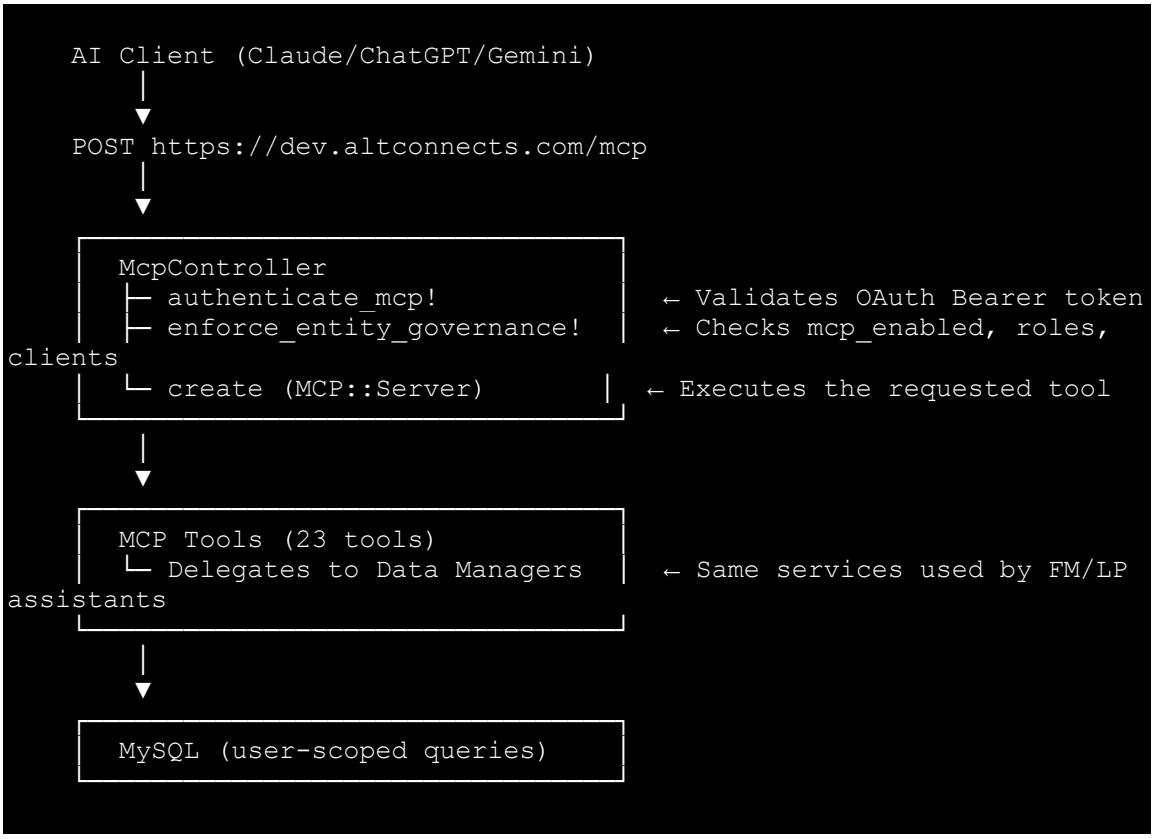
Should return JSON with `authorization_endpoint`, `token_endpoint`, etc.

Security Notes

- **Read-only access:** The MCP server only exposes read operations. No data can be created, modified, or deleted through MCP.
- **User-scoped:** Every query returns only data the authenticated user is authorized to see. No cross-tenant data is ever exposed.
- **OAuth tokens:** Tokens expire after 30 days. Users can revoke tokens from the CapHive UI.

- **Entity governance:** Admins control which entities, roles, and AI clients have MCP access.
- **All requests are logged:** Every MCP tool call is recorded in the server logs for audit purposes.

Architecture Overview (For Technical Team Members)



Transport: Streamable HTTP (single POST endpoint, stateless)
Auth: OAuth 2.1 with PKCE via Doorkeeper gem
Discovery: RFC 8414 + RFC 9728 `.well-known` metadata endpoints